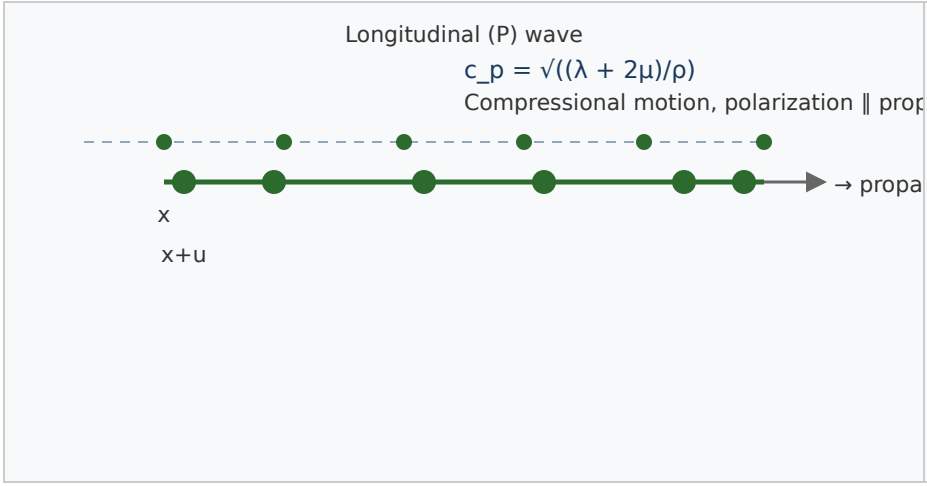
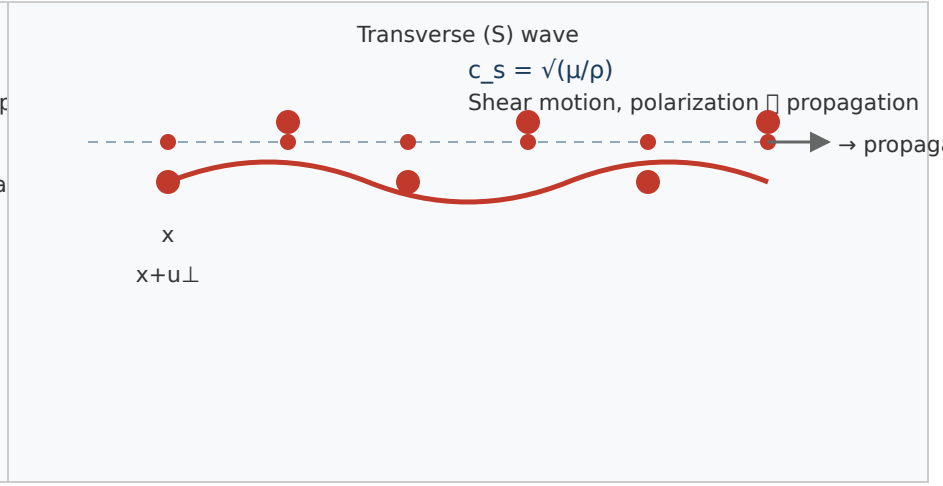
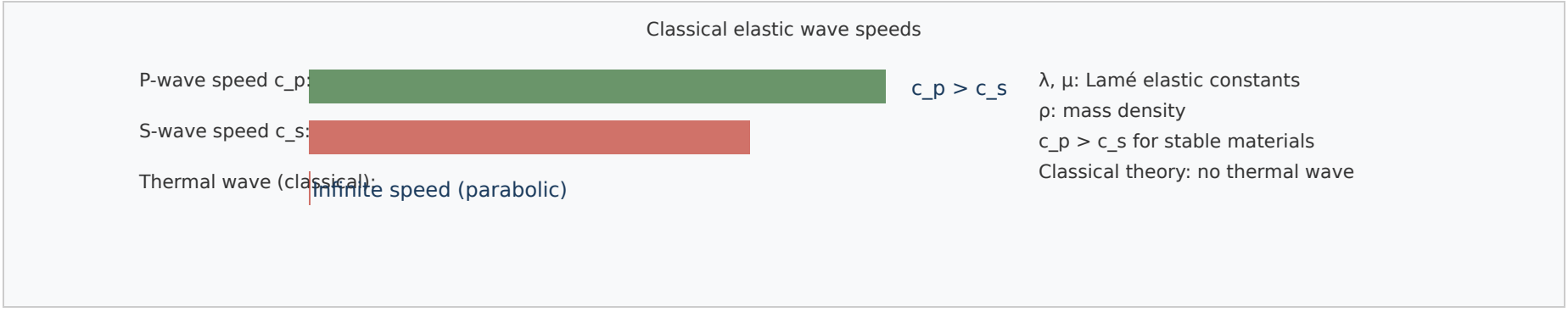
Classical elastic wave propagation

Longitudinal (P) wave and transverse (S) wave in an isotropic elastic solid







P-waves involve volume change (compressional); S-waves involve shear (distortional) deformation.
Both propagate without dispersion in classical elasticity. Thermal disturbances diffuse (infinite speed).